Shahyad Khamnei

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Education

Boston University, Boston, MA *Expected December 2027*
B.A. in Mathematics, Physics, & Computer Science
Graduate Coursework: Deep Learning Theory, Probability Theory, Optimization Algorithms, Quantum Computing

University of Geneva, Geneva, Switzerland *February – June 2026*
Spring 2026 Semester Exchange Student
Graduate Coursework: Measure Theory, Representation Theory & Lie Algebras

Research Experience

Max Planck Institute for the Physics of Complex Systems (MPIPKS), Dresden, Germany *Fall 2026*
Research Intern in Condensed Matter Divison –Nonequilibrium Quantum Dynamics (Advisor: Dr. Marin Bukov)
Project Title: TBD

CERN Theory Division, Geneva, Switzerland *Spring - Summer 2026*
Research Intern (Advisor: Dr. Kyriakos Papadodimas)
Project Title: Random Matrix Theory and Free Probability in Quantum Systems

Boston University Math & Statistics Department, Boston, MA *Fall 2025 – Fall 2026*
Undergraduate Researcher (Advisor: Prof. Konstantinos Spiliopoulos)
Project Title: Algorithms and Optimization of Random Vector Functional Link Neural Networks

- In collaboration with Los Alamos National Lab Applied Computer Science Group.

Boston University Condensed Matter Theory Group, Boston, MA *Summer - Fall 2025*
Undergraduate Researcher (Advisor: Prof. Anatoli Polkovnikov)
Project Title: Partial Reversibility and Counterdiabatic Driving in Nearly Integrable Systems

- Gave a talk presenting our results at American Physical Society 2026 Global Conference. Link Here
- ArXiv Preprint: Link Here

University of South Carolina REU, Columbia, SC *Summer 2024*
REU Participant (Group Advisor: Prof. Zhu Wang)
Project Title: Data-Driven Reduced Order Modeling of PDEs

Awards and Recognitions

Boston University Quantum Convergence Grant

- Used to fund my travel to the APS Global Summit 2026

Iranian-American Scholarship Foundation Scholar
Boston University Statistics Department Junior Award for Academic Excellence
Boston University Dean's List
North Charleston West Corridor Merit Scholar

Technical Skills

Technical Skills: Python (PyTorch, Pandas, NumPy, SymPy, MatPlotLib), C++ (Eigen), Java, Mathematica, Linux, Git Hub, Latex
Languages: English (Native Speaker), Farsi (Native Speaker), French (Basic - A2)

Coursework

Math & Statistics: Mathematical Theory of Deep Learning, Probability Theory, Measure Theory, Representation Theory & Lie Algebras, Financial Derivatives, Real Analysis I & II, Complex Analysis, Abstract Algebra I & II, Differential Equations, Linear Algebra, Multivariate Calculus
Physics: Quantum Physics I & II, Quantum Computing, Electricty & Magnetism, Computational Physics
Computer Science: Optimization Algorithms, Algorithms Analysis, Data Structures
Fall 2026 Semester: Functional Analysis, Topology, TBD

Leadership & University Engagement

Boston University PRISM Mentorship Program *2025 – 2026*
Student Advisor

- Advise first and second year students in math and physics regarding coursework, research, and their career paths.

Boston University Technology Inspiration Scholars Program *2024 – 2025*
Logistics Co-Director

- Organized a hackathon for underprivileged high school students in the Greater Boston area.

Boston University F.E.A.S.T. Branch of Student Government *2023 – 2025*
Treasurer

- F.E.A.S.T. stands for Food Equality, Accountability, Sustainability, and Transparency.
- Organized and Handled Funding for community service events and promoted food sustainability in the community and university.